

Camp 2310 - Lab 1 (unmarked)

- 1- Draw truth tables for the following expressions and identify if each one is a tautology, satisfiable, or contradiction.
 - (i) $p \wedge \sim (q \vee r)$
 - (ii) $\sim p \vee (q \vee r)$
- 2- Use truth table to prove if $\sim((p \vee h) \Rightarrow g)$ is equivalent to $\sim g \Rightarrow (p \vee h)$
- 3- Prove, by logical equivalences and inference rules that $(p \vee q) \wedge (r \vee p) \wedge (\sim q \vee \sim r \vee p)$ is logically equivalent to p .
- 4- Prove the following statement (distribution of implication over conjunction).
$$(p \Rightarrow (q \wedge r)) \Leftrightarrow ((p \Rightarrow q) \wedge (p \Rightarrow r))$$
- 5- prove that if $p \Rightarrow (q \wedge r)$ then $(\sim q \vee \sim r) \Rightarrow \sim p$.
- 6- prove by contradiction that if $(p \Rightarrow q) \wedge (r \Rightarrow s)$, $(q \Rightarrow t) \wedge (s \Rightarrow u)$, $\sim (t \wedge u)$, and $p \Rightarrow r$ then $\sim p$.

7- Three students X, Y, and Z are accused of cheating in an assignment. During the interrogation they made the following claims:

X says: Y cheated and Z did not cheat.

Y says: If X cheated then Z cheated too.

Z says: I did not cheat. One or maybe both of the others cheated.

(a) Represent above statements into propositional logic.

(b) Are the three statements contradicting each other or they are satisfiable?

(c) Assuming that nobody lied, who is innocent and who cheated?

8- During a murder investigation, we gathered the following clues:

(1) if the knife is in the store room, then we saw it when we cleared the store room.

(2) the murder was committed at the basement or inside the apartment.

(3) if the murder was committed in the basement then the knife is in the yellow dust bin.

(4) we did not see the knife when we cleared the store room.

(5) if the murder was committed outside the building, then we are unable to find the knife

(6) if the murder was committed inside the apartment, the the knife is in the store room

Use propositional logic to determine where the knife is.